# Kolmogorov-Smirnov goodness-of-fit tests

(page 63) :

**Data** : The data consists of a r.s $X_1, X_2, \ldots, X_n$ of size n associated with some unknown distribution function, denoted by $F(x)$.

**Assumptions**: 1) The sample is a random sample.
2) If the hypothesized distribution function, $F_0(x)$ in $H_0$, is continuous the test is exact. Otherwise the test is conservative.

**Hypotheses**: Let $F_0(x)$ be a completely specified hypothesized distribution function.

A. (Two-sided)
$H_0: F(x) = F_0(x)$ for all values of $x$
$H_1: F(x) \neq F_0(x)$ for at least one value of $x$.

B. (One-sided)
$H_0: F(x) \geq F_0(x)$ for all values of $x$
$H_1: F(x) < F_0(x)$ for at least one value of $x$

C. (One-sided)
$H_0: F(x) \leq F_0(x)$ for all values of $x$
$H_1: F(x) > F_0(x)$ for at least one value of $x$

**Test statistic**:

A. (Two-sided)

$$D_n = \sup_x |F_n(x) - F(x)|$$
$$= \max\left( F(x_i) - F_n(x_{i-1}),\ F_n(x_i) - F(x_i) \right)$$

B. (One-sided)

$$D_n^- = \sup_x [F(x_i) - F_n(x_{i-1})]$$

C. (One-sided)

$$D_n^+ = \sup_x [F_n(x_i) - F(x_i)]$$

**Decision Rule**:

Reject $H_0$ at $\alpha$ level of significance if the appropriate test statistic $D_n$, $D_n^+$ or $D_n^-$ $> D_n^\alpha$, where $D_n^\alpha$ is given by Table.